CSA1701 - ARTIFICIAL INTELLIGENCE

4.Question: Write a Python program to solve the Missionaries and Cannibals problem.

Aim: To transport three missionaries and three cannibals across a river without allowing cannibals to outnumber missionaries on either bank when missionaries are present.

Algorithm:

- Take the given crypt-arithmetic expression.
- Identify all unique letters.
- Assign different digits to each letter.
- Check the arithmetic expression.
- Repeat until a valid assignment is found.
- Print the solution.

Program:

5 python.py - C:/Users/praji/AppData/Local/Programs/Python/Python314/5 python.py (3.14.8)

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from collections import deque

start = (3, 3, 1)
goal = (0, 0, 0)
moves = [(1, 0), (2, 0), (0, 1),
         (0, 2), (1, 1)]

def valid(m, c):
    mr, cr = 3 - m, 3 - c
    return (0 <= m <= 3 and 0 <= c <= 3
            and (m == 0 or m >= c)
            and (mr == 0 or mr >= cr))

def successors(state):
    m, c, boat = state
    for dm, dc in moves:
        if boat == 1:
            nm, nc = m - dm, c - dc
        else:
            nm, nc = m + dm, c + dc

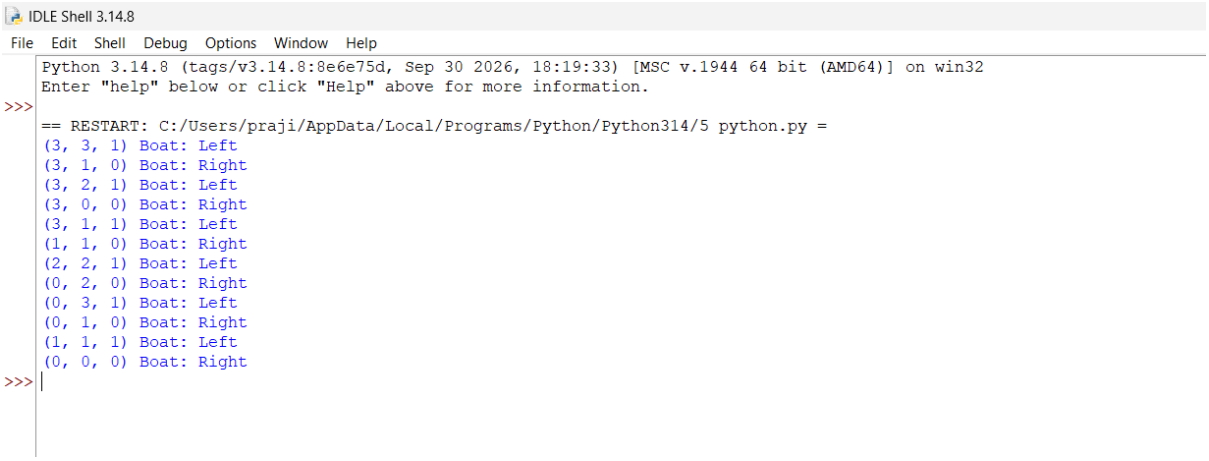
        if valid(nm, nc):
            yield (nm, nc, 1 - boat)

queue = deque([(start, [start])])
visited = {start}

while queue:
    state, path = queue.popleft()

    if state == goal:
        for s in path:
            print(s, "Boat:", "Left" if s[2] else "Right")
        break

    for nxt in successors(state):
        if nxt not in visited:
            visited.add(nxt)
            queue.append((nxt, path + [nxt]))
else:
    print("No solution")
```



Result: Thus, the Missionaries and Cannibals problem was solved using BFS while maintaining the safety constraints.